

Question No:5

DS	Session	Queue	Question Information	Level 1	Challenge 55
Question description					
Sathya is an DS expert training youngsters struggling in DS to make them better.					
Sathya usually gives interesting problems to the youngsters to make them love the DS.					
One such day Sathya provided to the youngsters to solve the task such that, insert an element in a Queue in FIFO order					
Youngsters were lacking the idea to solve the problem.					
Being an exciting youngster can you solve it?					
Function Description					

1. Define the maximum size of queue and initialize front and rear as -1.
2. In the main function we will initialize two variables that will store the data and the size of the queue.
3. Accept the data that we want to enter in a queue using a for loop.
4. After accepting the data use enqueue() function to insert the data in a queue.
5. In this function return queue is full , if the value of rear is equal to max-1.
6. Else increase the value of rear by 1.
7. After this insert the data that we have accepted earlier.
8. Now display the data of the queue using a disp() function.
9. In this function if the value of front is equal to -1 then it means list is empty.
10. Else initialize a for loop from front to rear and print the data that queue is holding.

Constraints:

$0 < \text{size} < 100$

$0 < \text{data} < 1000$

Input format:

First line indicates the size of the queue

Second line indicates the elements of the queue.

Output Format:

every lines indicates the enqueue of each elements

last line indicates the fine queued elements.

Logical Test Cases

Test Case 1

INPUT (STDIN)

5
5 6 3 1 2

EXPECTED OUTPUT

Enqueueing 5
5 Enqueueing 6
5 6 Enqueueing 3
5 6 3 Enqueueing 1
5 6 3 1 Enqueueing 2
5 6 3 1 2

Test Case 2

INPUT (STDIN)

8
1 8 6 4 9 5 3 5

EXPECTED OUTPUT

Enqueueing 1
1 Enqueueing 8
1 8 Enqueueing 6
1 8 6 Enqueueing 4
1 8 6 4 Enqueueing 9
1 8 6 4 9 Enqueueing 5
1 8 6 4 9 5 Enqueueing 3
1 8 6 4 9 5 3 Enqueueing 5
1 8 6 4 9 5 3 5

Mandatory Test Cases

Test Case 1

KEYWORD

void enqueue(int data)

Test Case 2

KEYWORD

for(i=front;i<=rear;i++)

Test Case 3

KEYWORD

enqueue(data);

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

Code:

```
1  #include <stdio.h>
2  #define MAX 1000
3
4  int queue[MAX];
5  int front = -1;
6  int rear = -1;
7
8  void enqueue(int data)
9  {
10     if (rear == MAX - 1)
11     {
12         return;
13     }
14
15     if (front == -1)
16     {
17         front = 0;
18     }
19
20     rear++;
21     queue[rear] = data;
22 }
23
24 void display()
25 {
26     for (int i = front; i <= rear; i++)
27     {
28         printf("%d", queue[i]);
29
30         if (i < rear)
31         {
32             printf(" ");
33         }
34     }
35 }
36
37 int main()
```

```

38 {
39     int n;
40     int data;
41
42     scanf("%d", &n);
43
44     for (int i = 0; i < n; i++)
45     {
46         scanf("%d", &data);
47
48         /* Display existing queue */
49         if (front != -1)
50         {
51             display();
52             printf(" ");
53         }
54
55         /* Show enqueueing message */
56         printf("Enqueueing %d\n", data);
57
58         /* Insert the element */
59         enqueue(data);
60     }
61
62     /* Display final queue */
63     display();
64
65     return 0;
66 }

```

Output:

Test Case-1:

```

5
5 6 3 1 2
Enqueueing 5
5 Enqueueing 6
5 6 Enqueueing 3
5 6 3 Enqueueing 1
5 6 3 1 Enqueueing 2
5 6 3 1 2
-----
Process exited after 15.59 seconds with return value 0
Press any key to continue . . .

```

Test Case-2:

```
8
1 8 6 4 9 5 3 5
Enqueueing 1
1 Enqueueing 8
1 8 Enqueueing 6
1 8 6 Enqueueing 4
1 8 6 4 Enqueueing 9
1 8 6 4 9 Enqueueing 5
1 8 6 4 9 5 Enqueueing 3
1 8 6 4 9 5 3 Enqueueing 5
1 8 6 4 9 5 3 5
-----
Process exited after 2.308 seconds with return value 0
Press any key to continue . . . |
```

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